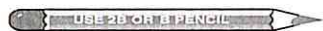


HOW TO FILL OUT THIS SHEET:



- Rub out all mistakes completely.
- Print your details clearly in the boxes provided.
- Make sure you fill in only one oval in each column.

EXAMPLE 1: Debbie Bach

FIRST NAME

LAST NAME

DEBBIE

D	E	B	B	I	E
A	A	A	A	A	A
B	B			B	B
C	C	C	C	C	
C	B	B	B		
E		E			

BACH

B	A	C	H	
A		A	A	
	B	B	B	
C	C			
B				
E				

EXAMPLE 2: Chan Ai Beng

FIRST NAME

LAST NAME

Two 5x5 grids for the word 'CHAN' and 'AIBENG'. The first grid is for 'CHAN' and the second is for 'AIBENG'. Each grid has a header row with the letters of the word. Below the header row are four rows of circles, each containing a letter. The letters are arranged in a 4x5 grid. The letters are: Row 1: A, A, A, A, A; Row 2: B, B, B, B, B; Row 3: C, C, C, C, C; Row 4: D, D, D, D, D; Row 5: E, E, E, E, E. The letters are arranged in a 4x5 grid. The letters are: Row 1: A, A, A, A, A; Row 2: B, B, B, B, B; Row 3: C, C, C, C, C; Row 4: D, D, D, D, D; Row 5: E, E, E, E, E.

EXAMPLE 3: Jamal bin Abas

FIRST NAME

LAST NAME

[illegible]

FIRST NAME to appear on certificate

[illegible]

LAST NAME to appear on certificate

[illegible]

Are you male or female?

☐ Male ☐ Female



Does anyone in your home usually speak a language other than English?

☐ Yes ☐ No

School name: _____

Town / suburb: _____

Today's date: / / Postcode:

DATE OF BIRTH

Day		Month	Year	
0	0	0	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7	7	7	7	7
8	8	8	8	8
9	9	9	9	9

CLASS
(optional)

(A)	(K)
(B)	(L)
(C)	(M)
(D)	(N)
(E)	(O)
(F)	(P)
(G)	(Q)
(H)	(R)
(I)	(S)
(J)	(T)

TO ANSWER THE QUESTIONS

MULTIPLE CHOICE

Questions 1 to 40

Example: $4 + 6 =$

- (A) 2
- (B) 9
- (C) 10
- (D) 24

The answer is 10, so fill in the oval (C), as shown.

(A) (B) ☒ (C) (D)



START

- 1 (A) (B) (C) (D)
- 2 (A) (B) (C) (D)
- 3 (A) (B) (C) (D)
- 4 (A) (B) (C) (D)
- 5 (A) (B) (C) (D)
- 6 (A) (B) (C) (D)
- 7 (A) (B) (C) (D)
- 8 (A) (B) (C) (D)
- 9 (A) (B) (C) (D)
- 10 (A) (B) (C) (D)

- 11 (A) (B) (C) (D)
- 12 (A) (B) (C) (D)
- 13 (A) (B) (C) (D)
- 14 (A) (B) (C) (D)
- 15 (A) (B) (C) (D)
- 16 (A) (B) (C) (D)
- 17 (A) (B) (C) (D)
- 18 (A) (B) (C) (D)
- 19 (A) (B) (C) (D)
- 20 (A) (B) (C) (D)

- 21 (A) (B) (C) (D)
- 22 (A) (B) (C) (D)
- 23 (A) (B) (C) (D)
- 24 (A) (B) (C) (D)
- 25 (A) (B) (C) (D)
- 26 (A) (B) (C) (D)
- 27 (A) (B) (C) (D)
- 28 (A) (B) (C) (D)
- 29 (A) (B) (C) (D)
- 30 (A) (B) (C) (D)

- 31 (A) (B) (C) (D)
- 32 (A) (B) (C) (D)
- 33 (A) (B) (C) (D)
- 34 (A) (B) (C) (D)
- 35 (A) (B) (C) (D)
- 36 (A) (B) (C) (D)
- 37 (A) (B) (C) (D)
- 38 (A) (B) (C) (D)
- 39 (A) (B) (C) (D)
- 40 (A) (B) (C) (D)



M

2011 Mathematics Answer Keys

ICAS
International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
1	B	A	B	B	A	A	C	A
2	B	D	C	D	B	D	B	D
3	D	B	D	C	B	D	A	C
4	D	A	B	C	C	D	D	A
5	C	A	A	B	C	A	A	D
6	C	B	C	D	B	A	C	A
7	A	C	D	A	A	B	A	B
8	D	D	C	D	B	D	C	B
9	D	B	D	A	D	A	D	C
10	A	B	D	A	D	B	A	C
11	C	D	A	D	D	D	D	A
12	D	C	A	D	C	A	B	A
13	C	B	B	C	B	A	A	C
14	C	C	B	D	A	A	B	B
15	A	B	B	A	A	D	D	C
16	B	D	A	A	D	B	A	B
17	D	A	A	B	A	C	B	D
18	A	D	D	A	D	C	C	D
19	B	A	D	A	C	C	B	A
20	B	C	A	C	C	B	B	C
21	A	D	D	B	D	A	A	C
22	C	D	C	B	B	C	C	C
23	D	B	A	C	C	B	B	B
24	D	D	C	C	B	A	C	D
25	A	A	C	B	D	B	B	B
26	A	B	D	B	B	B	B	C
27	C	A	A	C	A	B	C	B
28	A	D	B	C	D	C	A	B
29	A	D	B	B	A	B	A	D
30	B	C	D	D	D	C	C	B

(Please turn over)



M

2011 Mathematics Answer Keys

ICAS
International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
31	C	C	B	B	C	D	C	C
32	B	C	C	D	B	C	D	B
33	C	A	C	A	C	D	D	C
34	B	B	C	C	D	C	A	D
35	B	A	A	C	C	B	B	A
36	A	C	C	21 021	900	56 056	108	3 03 003 0 3
37	C	B	D	43 043	600	80 080	3 03 003 0 3	76 076
38	C	A	C	25 025	39 039	28 028	83 083	439
39	D	C	D	40 040	75 075	114	9 09 009 0 9	947
40	B	D	B	8 08 008 0 8	131	7 07 007 0 7	901	784

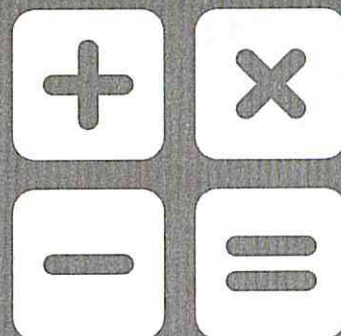
Contact Details

Telephone: (02) 8344 1016
Email: info@eaa.unsw.edu.au
Website: www.eaa.unsw.edu.au



UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

**PAPER
B**



2011
ICAS

International Competitions
and Assessments for Schools

MATHEMATICS

**Educational
Assessment
Australia**
eaa.unsw.edu.au

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.

You are **NOT** allowed to use a calculator.

1. This diagram shows the locations of some activities at Fun Park.



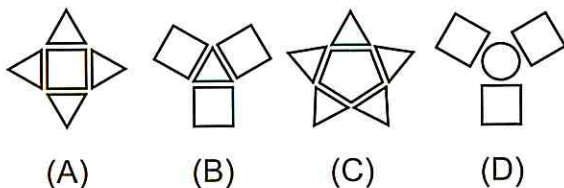
Which activity is shown at B2?

- (A) Go-Karts
- (B) Row Boats
- (C) Roller Coaster
- (D) Adventure Golf

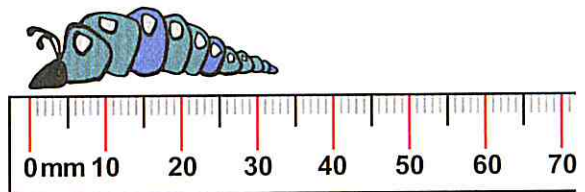
2. Tony made a design that contains this shape.



Which of these could **NOT** be Tony's design?



3. Anish found a caterpillar.



What is the length of this caterpillar?

- (A) 30 mm
- (B) 33 mm
- (C) 30 cm
- (D) 33 cm

4. The houses on one side of Station Street have odd numbers. Each block of land has one house on it.

The picture shows the first five houses in Station Street.



Tony lives at number 19 Station Street.
Yara lives at number 29 Station Street.

How many houses are there between Tony's house and Yara's house?

- (A) 4
- (B) 5
- (C) 9
- (D) 10

5. The symbol  stands for the number of pets Anish has.

Tony has three more pets than Anish.

How many pets does Tony have?

- (A)  + 3
(B)  - 3
(C) 3 - 
(D) 3 × 

6. Natalia is making cupcakes. She needs 60 g of butter to make 12 cupcakes.



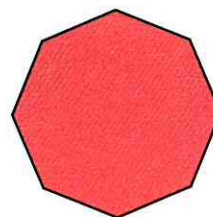
What is the largest number of cupcakes that Natalia can make with 120 g of butter?

- (A) 10
(B) 24
(C) 60
(D) 132

7. A square has two pairs of parallel sides.



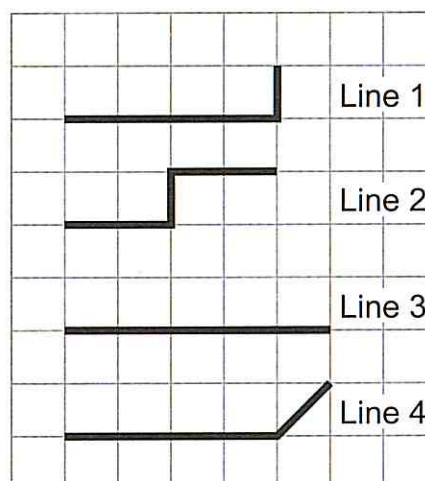
How many pairs of parallel sides does a regular octagon have?



- (A) 0
(B) 2
(C) 4
(D) 8

8. There are four black lines drawn on the grid.

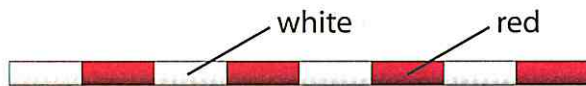
They are labelled Line 1, Line 2, Line 3 and Line 4.



Which black line is the longest?

- (A) Line 1
(B) Line 2
(C) Line 3
(D) Line 4

9. A stick is painted with equal strips of red and white.



Which picture shows half of this stick?



(A)



(B)



(C)

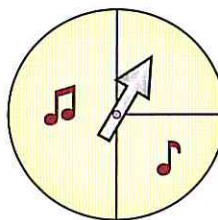


(D)

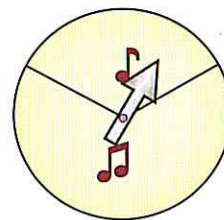
10. $79 + 52 + 88 = ?$

- (A) 209
- (B) 219
- (C) 2019
- (D) 2119

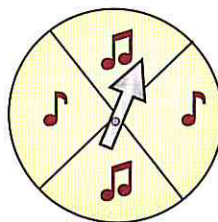
11. Which spinner has the least chance of landing on  ?



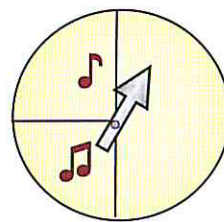
(A)



(B)



(C)



(D)

12. Tony left home at 8:30 am to go on a day trip. He arrived back home at 3:40 pm.



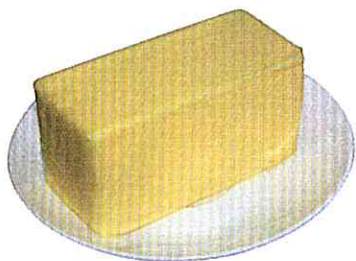
How long was Tony away from home?

- (A) 4 hours 50 minutes
- (B) 5 hours 10 minutes
- (C) 7 hours 10 minutes
- (D) 11 hours 70 minutes

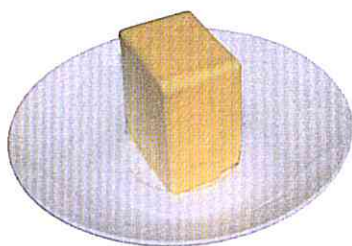
13. Which of these is the longest?

- (A) 1 month
- (B) 6 weeks
- (C) 27 days
- (D) 48 hours

14. This block of butter has a mass of 250 grams.



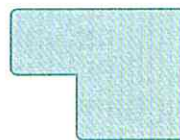
Yara needs to use the amount of butter shown below for a biscuit recipe.



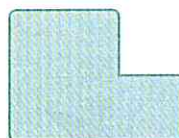
About how many grams of butter does Yara need?

- (A) 50
- (B) 65
- (C) 85
- (D) 125

15. Anish drew this shape.



Which option shows the shape after a quarter turn clockwise?



(A)



(B)



(C)



(D)

16. Jess and her friends were playing 'Pick Up Sticks'.

This table shows the points scored for picking up different coloured sticks.

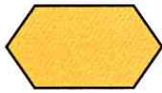
Colour of stick	Point value
Yellow	1
Green	2
Blue	5
Red	10
Black	25

Jess picked up five yellow, five green, five blue and five red sticks.

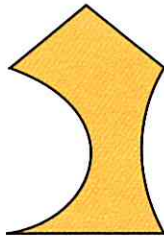
What was the total point value of the sticks that Jess picked up?

- (A) 18
- (B) 20
- (C) 43
- (D) 90

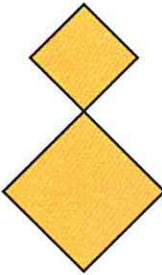
17. Natalia drew four shapes.



Shape 1



Shape 2



Shape 3

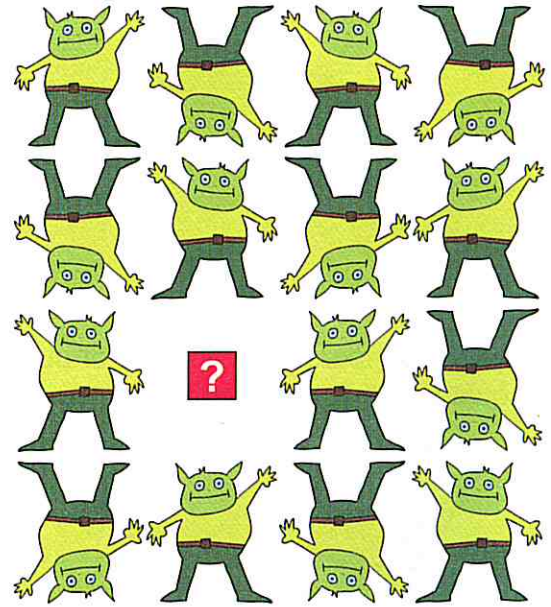


Shape 4

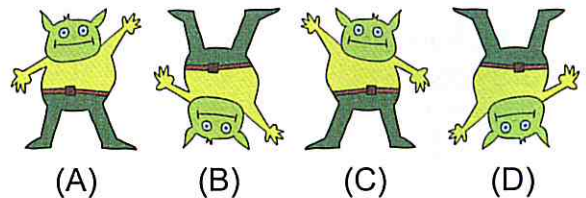
Which shapes have only one line of symmetry?

- (A) Shape 3 and Shape 4
- (B) Shape 2 and Shape 3
- (C) Shape 1 and Shape 3
- (D) Shape 1 and Shape 2

18. A picture is missing from this pattern.



Which picture is missing?



19. A school has 150 students. Two-thirds of the students are boys.

How many students are girls?

- (A) 50
- (B) 75
- (C) 100
- (D) 125

20. At a party, four flavours of milkshake were served. Each child was allowed one milkshake.

Flavour	Number of children
Banana	4
Chocolate	7
Strawberry	10
Vanilla	10

How many children chose banana or chocolate?

- (A) 4
(B) 7
(C) 11
(D) 20

21. There are 50 students in Year 3.

The table shows the number of Year 3 boys and girls in each sport group. Each student is in only one group.

Sport group	Girls	Boys
Red	3	9
Blue	8	7
Green	11	?

How many boys are in the Green group?

- (A) 38
(B) 34
(C) 16
(D) 12

22. A local shop is offering one bouncy ball free with every five bouncy balls a customer pays for.

Jess pays the normal price for 30 bouncy balls.

How many bouncy balls will she receive in total?

- (A) 30
(B) 31
(C) 35
(D) 36

23. Jess wrote this number sentence

$$2 \times \text{😊} - \text{😊} = 2 \times \text{★} + \text{😊}$$

Which option makes this number sentence true?

- (A) 😊 = 1, ★ = 1
(B) 😊 = 2, ★ = 0
(C) 😊 = 4, ★ = 3
(D) 😊 = 6, ★ = 2

24. Anish cut out these 12 cards and put them in a bag.

★	😊	🌸	🌸
↗	🌸	★	↗
↗	♥	😊	↗

Anish picked a card from the bag without looking.

Which two pictures have the same chance of being on the card Anish picked?

- (A) ↗ and 🌸
(B) ↗ and ★
(C) ★ and 🌸
(D) ★ and 😊

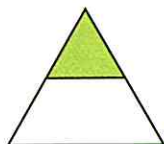
25. Which of these 2D shapes does NOT have one quarter coloured in?



(A)



(B)



(C)

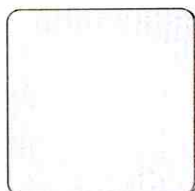


(D)

26. Shadia has two kinds of square tiles. The small tiles are 20 cm long. The big tiles are 30 cm long.



20 cm



30 cm

She makes a line of small tiles that is 180 cm long. She also makes a line of big tiles that is 120 cm long.

How many more tiles are in the line that is 180 cm long?

- | | |
|--------|-------|
| (A) 60 | (B) 5 |
| (C) 3 | (D) 2 |

27. Jess started listening to a CD when the minute hand on the clock was pointing to the 9.

When she finished listening to the CD, the minute hand on the clock was pointing to the 5.

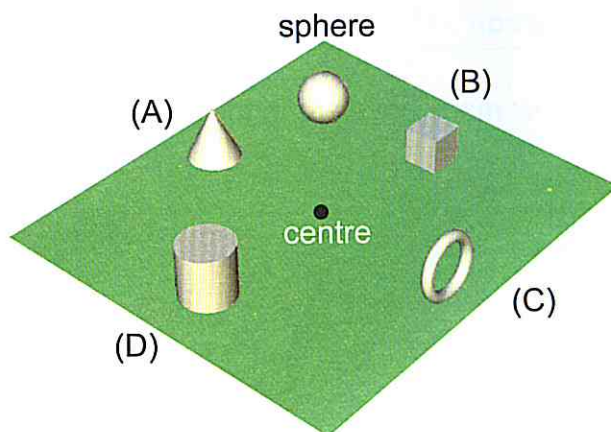


How long could Jess have listened to the CD?

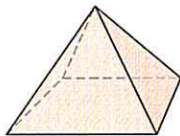
- | |
|----------------|
| (A) 40 minutes |
| (B) 20 minutes |
| (C) 14 minutes |
| (D) 4 minutes |

28. The picture shows five sculptures in a park. The sphere is south-east of the centre.

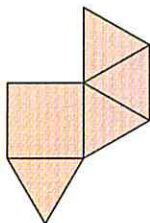
Which sculpture is north of the centre?



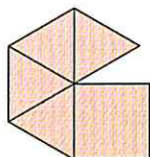
29. This is a square pyramid.



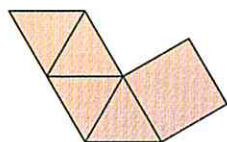
Which option shows the net of a square pyramid?



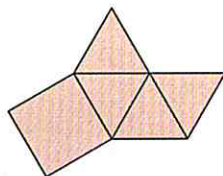
(A)



(B)



(C)



(D)

30. Sally wrote a story about two kinds of monsters.

Sally's monsters



1 head
3 eyes

2 heads
4 eyes

In the story there are 12 monsters.
Altogether they have 46 eyes.

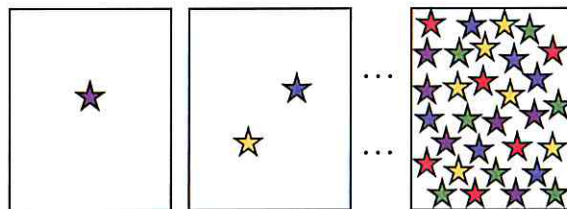
How many two-headed monsters are in the story?

- (A) 2
(B) 6
(C) 10
(D) 11

31. Natalia had a sticker book with 31 pages.

On the 1st page she put one sticker.
On the 2nd page she put two stickers.

She continued this pattern to the 31st page.



1st page

2nd page

31st page

How many stickers did Natalia put in her book altogether?

- (A) 196
(B) 306
(C) 496
(D) 961

32. Anish traded all his old video games for new ones.



The game store gave Anish one new video game for every two old ones.

Then he was given four more new video games for his birthday.

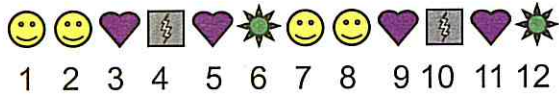
Now Anish has ten new video games.

How many old video games did Anish have?

- (A) 3
(B) 7
(C) 12
(D) 16

33. Jess arranged some shapes into a repeating pattern.

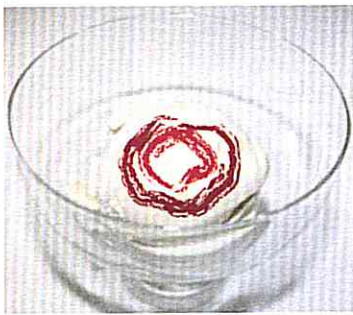
Here are the first 12 shapes in the pattern.



What will be the 46th shape in this pattern?

- (A) 
 (B) 
 (C) 
 (D) 

34. Jess makes a scoop of ice-cream that has a volume of 85 mL.

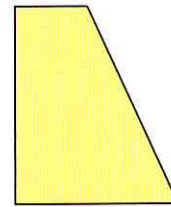


What is the largest number of scoops of this size that Jess could make from a 2 L tub of ice-cream?

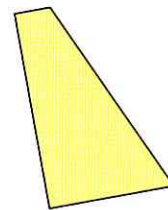
- (A) 2
 (B) 23
 (C) 24
 (D) 42

35. Tony cut a square into two pieces.

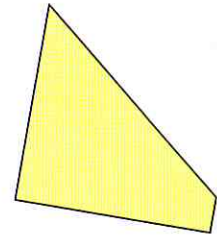
This is one of the pieces.



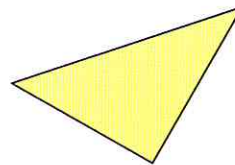
Which of these is the other piece?



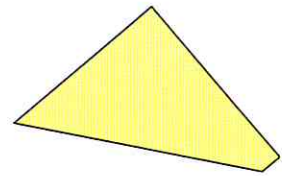
(A)



(B)

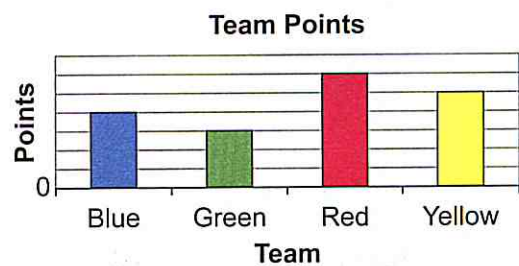


(C)



(D)

36. This graph shows the points scored by four teams.



The Yellow team has 10 more points than the Green team.

How many points does the Red team have?

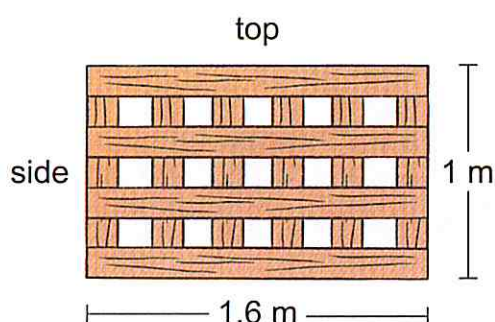
- (A) 6
 (B) 15
 (C) 30
 (D) 60

37. The larger of two numbers is four more than three times the smaller number. The sum of the two numbers is 28.

What is the smaller number?

- (A) 4
- (B) 6
- (C) 7
- (D) 8

38. Tony wants to put a fence around his yard using fence panels, like the one shown in the diagram. Each fence panel costs \$12.

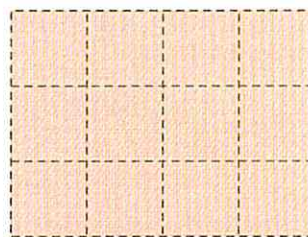


Tony's yard is square-shaped and has an area of 64 m^2 .

How much will it cost Tony to put a fence around his yard?

- (A) \$240
- (B) \$384
- (C) \$480
- (D) \$1920

39. The diagram shows a grid of squares.

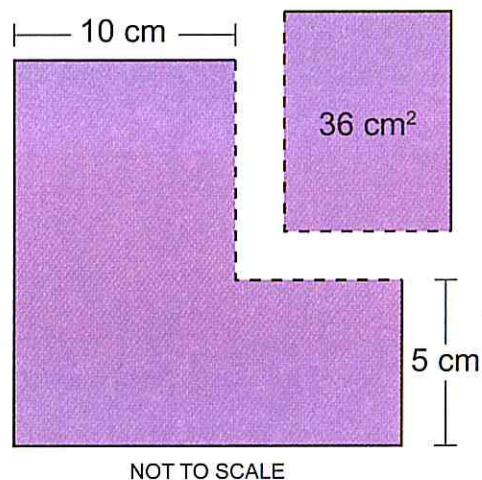


How many squares of any size can be traced along the dotted lines?

- (A) 12
- (B) 18
- (C) 20
- (D) 30

40. Natalia had a square piece of paper. She cut a rectangle out of the square as shown.

The area of the rectangle is 36 cm^2 .



What is the area of Natalia's original square, in cm^2 ?

- (A) 86
- (B) 144
- (C) 176
- (D) 196

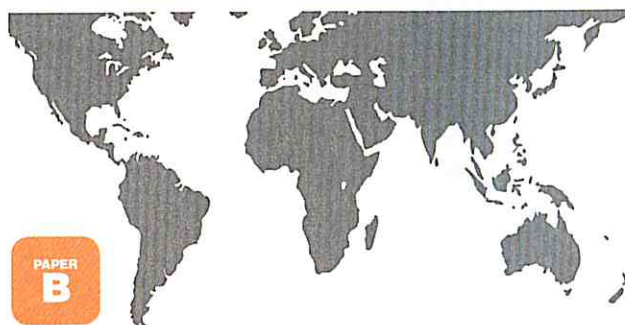
Acknowledgment

Copyright in this booklet is owned by Educational Assessment Australia, UNSW Global Pty Limited, unless otherwise indicated. Every effort has been made to trace and acknowledge copyright. Educational Assessment Australia apologises for any accidental infringement and welcomes information to redress the situation.



The following year levels should sit **THIS Paper**:

Australia	Year 4
Brunei	Primary 4
Indonesia	Year 5
Malaysia	Standard 4
New Zealand	Year 5
Pacific	Year 4
Singapore	Primary 3
South Africa	Grade 4



UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

**Educational
Assessment
Australia**
eaa.unsw.edu.au

© 2011 Educational Assessment Australia. EAA is an education group of UNSW Global Pty Limited, a not-for-profit provider of education, training and consulting services and a wholly owned enterprise of the University of New South Wales. ABN 62 086 418 582